DATA INTEGRITY HANDBOOK

The Growth Engineer’s Data Hygiene Guide

Bad telemetry data produces bad growth decisions. The 6-step quarterly checklist to audit, clean, and protect your event pipelines.

WHAT THIS AUDIT COVERS:

- Schema strictness at the API boundary to prevent corrupted metrics
- Orphan event pruning to reduce cloud analytics spend
- Mandatory UTC timestamp normalization for accurate Day-1 cohorts
- Programmatic lowercase UTM taxonomies and CSV formula injection defense

AUDIT 1 & 2: SCHEMAS & ORPHANS

Edge Validation & Orphan Events

Two structural audits to eliminate corrupt or unnecessary data before it enters your warehouse:

Audit 1: Schema Enforcement at the Edge

Problem: Flexible JSON schemas sound convenient until frontend sends strings where integers are expected.
Fix: Enforce strict Pydantic validation at the gateway. Quarantine malformed payloads before they reach DB.

Audit 2: Orphan Event Pruning

Problem: Deprecated features from 2024 still fire 50,000 events a day, inflating cloud analytics storage costs.
Fix: Query events with zero dashboard queries over 90 days. Safely sunset unused telemetry hooks.

AUDIT 3 & 4: TIMESTAMPS & DEDUPE

UTC Normalization & Idempotency

Ensure cohort metrics and retention curves reflect true user activity across global timezones:

Audit 3: Mandatory Server-Side UTC

Problem: Client device clocks are frequently inaccurate or in local time, distorting Day-1 retention cohorts.

Fix: Assign ISO-8601 UTC timestamps at server ingestion time. Never trust client device clocks for ordering.

Audit 4: Idempotent Event Deduplication

Problem: Mobile network reconnects resend HTTP requests, double-counting signup conversions.

Fix: Generate deterministic SHA-256 fingerprints and enforce unique constraints in the analytics database.

AUDIT 5 & 6: UTMS & SECURITY

UTM Taxonomies & Injection Defense

Protect your attribution reporting integrity and downstream spreadsheet consumers:

Audit 5: Programmatic Lowercase UTMs

Problem: Inconsistent casing like utm_source=LinkedIn vs linkedin splits attribution into disjoint buckets.

Fix: Enforce strict lowercase transforms and validated platform enums in all programmatic link builders.

Audit 6: CSV Formula Injection Protection

Problem: Telemetry values starting with =, +, -, or @ can execute arbitrary formulas in spreadsheet applications.

Fix: Sanitize exports by prefixing potentially dangerous leading characters with a single quote.

DATA HEALTH COCKPIT

Automated Telemetry Health Monitoring

Configure automated alerts to catch pipeline anomalies before stakeholders notice:

CRITICAL ALERT THRESHOLDS

- Event Volume Drops: Alert if hourly volume falls > 30% below 7-day rolling average
- Schema Rejection Rate: Alert if malformed payload rate exceeds 0.5% of total traffic
- Ingestion Latency: Alert if P99 edge gateway response exceeds 50ms
- Queue Backlog: Alert if consumer message queue exceeds 10,000 pending events

QUARTERLY HYGIENE

Run Your Quarterly Data Audit

Clean data is a competitive advantage. Instrument pipelines that engineering and leadership can trust implicitly.

DOWNLOAD THE DATA HYGIENE GUIDE

Get our comprehensive 6-step checklist, SQL audit queries, and CSV sanitization snippets via the verified link in this post.